



EPIC Mower: Weatherproof Landscaping Business

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EPIC Mower : Landscaping Business Challenges



► Company Overview

- EPIC Mower is a leading commercial landscaping company with a workforce of 250 employees, servicing hundreds of clients across the Bay Area. Our services include lawn maintenance, tree services, irrigation installation and repair, landscaping design, and installation. Many of these services are weather-sensitive, and schedule disruptions can lead to delays, revenue loss, and decreased customer satisfaction.

► Weather Impact on Services

- In addition to temperature and precipitation, other weather conditions such as wind speed, wind gusts, and soil moisture significantly impact lawn and tree services. Currently, EPIC Mower relies on branch managers to manually check weather information and adjust schedules based on their experience. However, the weather at the branch office may not accurately represent conditions at the service of appointment.

► Current Tools and Challenges

- The existing field service tools excel in scheduling, assigning qualified resources, routing, and estimating travel time. However, the lack of real-time, precise weather forecasts for service appointment locations is a major inefficiency. This gap in the system leads to suboptimal scheduling and resource allocation, ultimately affecting our operational efficiency and customer satisfaction.

- **Solution :** EPIC Mower needs to integrate real-time weather forecasting capabilities into the field service tools. Additionally, leveraging collective knowledge from SMEs to evaluate weather risk of each appointment will empower branch manager to make informed scheduling decisions. This enhancement will improve service delivery and maintain high levels of customer satisfaction.

EPIC Mover : Weatherproof Field Service Operation



1. In this Agentforce solution, we use Salesforce Field Service integrate with Open Meteo for real time hourly weather forecast at appointment location and time.
2. Agentforce uses Retrieval-augmented generation (RAG) to process unstructured documents uploaded to the Data Library by subject matter experts (SMEs). This process identifies high-risk weather criteria and thresholds for various landscaping services provided by EPIC Mower.
3. Agentforce Topic instructions guide the LLM on how to send precise geocode locations, start and end times of appointments to Open Meteo.
4. These instructions also direct the LLM to process the hourly weather forecast responses and display them in a user-friendly manner.
5. Subsequent Topic instructions enable the LLM to evaluate the weather risks associated with landscaping work and the conditions at the scheduled time. The risk criteria and thresholds are derived from the documents in the Data Library.
6. Of course Agentforce also instructed to find better time to reschedule the affected appointment and send email notification to customer.

How it works : Demo



1. Weather Forecast Inquiry:

- ▶ Prompt: "What's the weather forecast for San Francisco in the next 7 days?"
- ▶ Response: Agentforce retrieves the 7-day weather forecast for San Francisco from Open Meteo.
- ▶ EPIC Mower: Assess which day likely has service disruption by weather.

2. High-Risk Appointment Evaluation:

- ▶ Prompt: "Do you see any service appointments with high risk on 5/3?"
- ▶ Response: Agentforce evaluates the weather risk for appointment scheduled on May 3rd and identifies any high-risk appointments based on criteria derive from Agentforce data library.
- ▶ EPIC Mower: Validated service appointment at risk (Human in the loop). And ask for new schedule for SA-0276.

3. Low-Risk Scheduling:

- ▶ Prompt: "Please retrieve the weather forecast and find out which day before the due date of SA-0276 has low risk for tree pruning."
- ▶ Action: Agentforce retrieve and analyzes the weather forecast to determine the best low-risk day for tree pruning before the due date of service appointment SA-0276.
- ▶ EPIC Mower: Accept and ask to send email notification.

4. Customer Communication:

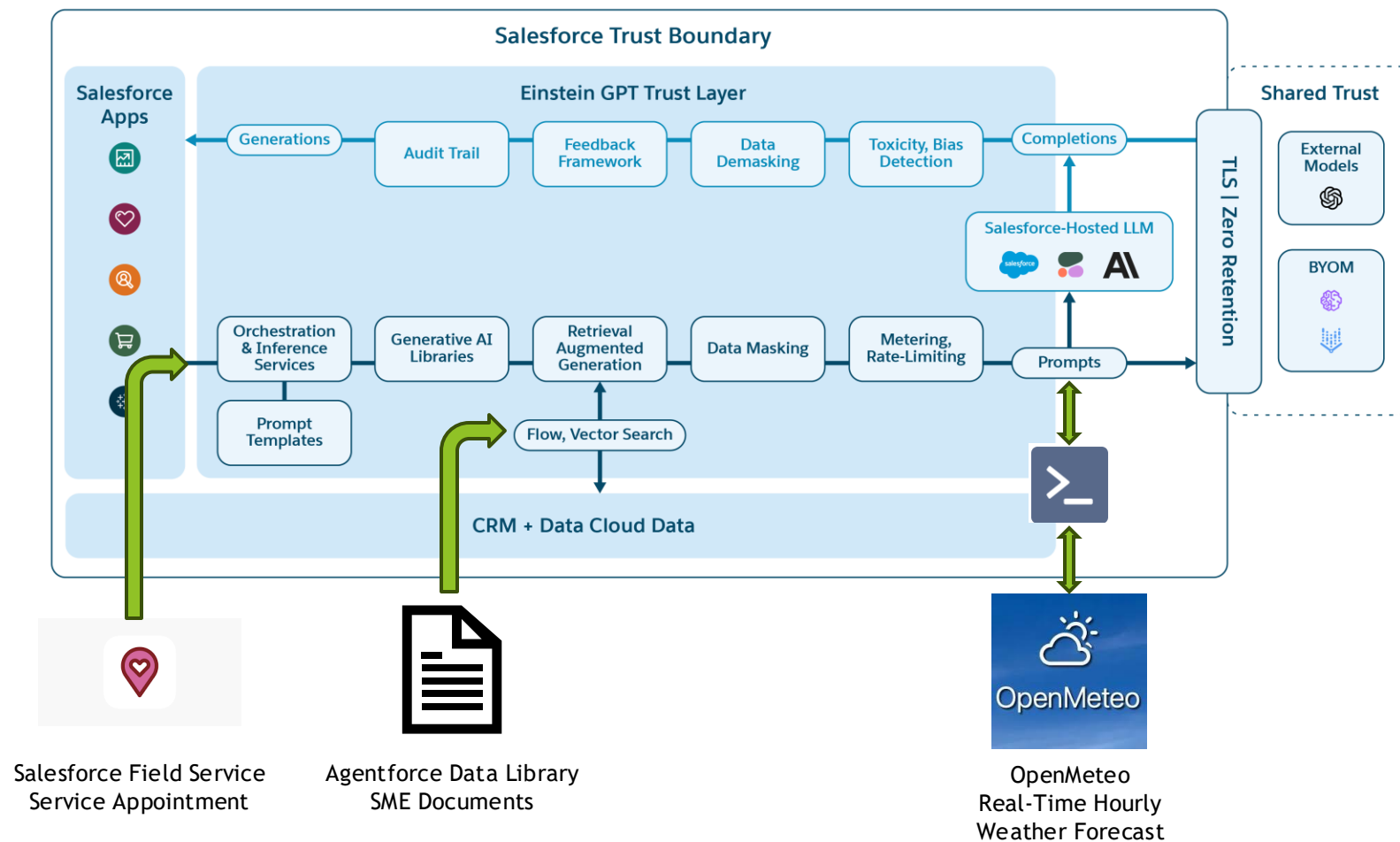
- ▶ Prompt: "Please proceed and draft an email to explain it to the customer politely."
- ▶ Action: Agentforce update the appointment and generates a polite and informative email to the customer, explaining the rescheduling due to weather conditions.

How we built it : Design Decision



- ▶ **We advocate for an elegant and efficient less is more design principle in this project.**
- ▶ **Real-Time Data Retrieval:** We eliminate conventional batch processing for storing weather data. With Agentive AI, we retrieve only the relevant data at the right time. We get this done with a single APEX class.
- ▶ **Dynamic Criteria Matching:** Instead of using predefined work types and weather risk criteria, Agentforce Data Library grounds the AI to determine the best match criteria and thresholds based on the appointment's work description. This approach avoids the need to customize existing data models or create custom objects.
- ▶ **Scalable and Lightweight Solution:** This solution is lightweight and scalable, allowing easy deployment to new weather-sensitive industries. Simply upload industry-specific SME documents to get started.

How we built it : Technologies used



If we have more time



- ▶ We could have fine tune the Agentforce Topic Instruction or customized the Model to get more consistent output.
- ▶ To support the use case of service resource team (crew) which is what happen in real world scenario
- ▶ Setup a demo to get feedback from my contact who is a manager of landscaping company.
- ▶ Integrate with Slack, send Slack notification to service resources in the area with unexpected severe weather condition.
- ▶ Generate the field service work rules from the Agentforce Data Library. This will effectively integrate this solution into Salesforce Field Service's robust scheduling engine.
- ▶ Team up with elite developer.
- ▶ Redo voice over of demo video.